

STATSIG lecture 9

Spectral estimation =
how to make robust PSD estimates from data
Parametric (model-based/grey-box) methods
(previous lecture non-parametric methods)

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Parametric (model based) estimation of PSD

For long signal blocks, e.g. 1024 samples (good possibilities for doing averaging of subblocks), the classical non-parametric methods for PSD estimation such as Welch's method or Blackman-Tukey works well and is de facto standard.

For short signal blocks, however, the parametric methods offer improvement

Overall parametric PSD estimation methodology (details on blackboard):

Assume the available signal block to have been generated by LTID-filtering of white noise
Estimate the transfer function $H(z)$, $H(f)$ of the hypothetical LTID-system
Estimate PSD from the LTID-system estimate

LTID-system types

Only poles in $H(z)$: autoregressive (AR) model (best for finding peaks in PSD, most used model)

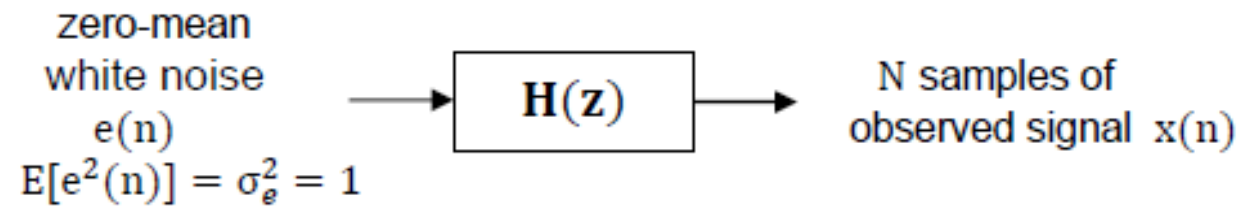
Only zeroes in $H(z)$: moving average (MA) model (best for smooth spectra, not so often used)

Both poles and zeroes in $H(z)$: ARMA model, most general, but harder mathematically

System order

Number of poles/zeroes must be assumed or iterated

AR(p) model: autoregressive model with p poles



AR(p) (autoregressive) :

$$x(n) + \sum_{i=1}^p a_i x(n-i) = b_0 e(n)$$
$$H(z) = \frac{b_0}{1 + \sum_{i=1}^p a_i z^{-i}}$$

General method

- Given N samples of observed signal $x(n)$
- Estimate $\hat{r}_x(k) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) x(n+k)$
- Assume orders (p, q) (# of poles / zeroes)
- Using $\hat{r}_x(k)$, find $\{\hat{a}_i\}_{i=1,\dots,p}$, $\{\hat{b}_i\}_{i=1,\dots,q}$ from relationship $\{a_i\}_{i=1,\dots,p}$, $\{b_i\}_{i=1,\dots,q} \rightarrow r_x(k)$
- $\{\hat{a}_i\}_{i=1,\dots,p}$ and $\{\hat{b}_i\}_{i=1,\dots,q}$ gives $\hat{H}(z)$ which gives $\hat{H}(f)$
- Then $\hat{P}_x(f) = \sigma_e^2 |\hat{H}(f)|^2 = |\hat{H}(f)|^2$

Finding LTID-system coefficients from autocorrelation estimate (details on blackboard)

Yule-Walker equations:
$$\begin{bmatrix} \hat{r}_x(0) & \cdots & \hat{r}_x(p-1) \\ \vdots & \ddots & \vdots \\ \hat{r}_x(p-1) & \cdots & \hat{r}_x(0) \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \vdots \\ \hat{a}_p \end{bmatrix} = - \begin{bmatrix} \hat{r}_x(1) \\ \vdots \\ \hat{r}_x(p) \end{bmatrix}, \quad \hat{b}_0^2 = \hat{r}_x(0) + \sum_{k=1}^p \hat{a}_k \hat{r}_x(k)$$

Solution of Yule-Walker equations:

- YW-method: solve equations directly, MATLAB: *aryule* $\rightarrow \{a_i\}$, *pyulear* $\rightarrow \hat{P}_x(f)$
- Covariance method, MATLAB: *arcov* $\rightarrow \{a_i\}$, *pcov* $\rightarrow \hat{P}_x(f)$
- Modified covariance method, MATLAB: *armcov* $\rightarrow \{a_i\}$, *pmcov* $\rightarrow \hat{P}_x(f)$
- Burg method, MATLAB: *arburg* $\rightarrow \{a_i\}$, *pburg* $\rightarrow \hat{P}_x(f)$

Guidelines for choosing model order p (number of poles in LTID)

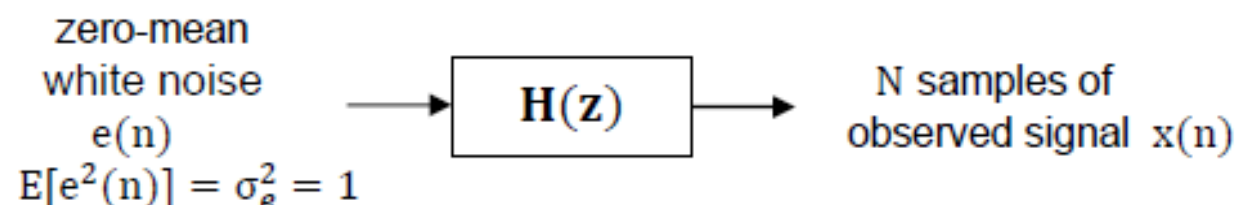
p is "too big": spurious peaks appear in PSD estimate (overfitting)

p is "too small": PSD estimate will be overly smoothed (poor resolution, underfitting)

Plot model "error power" $\hat{\varepsilon}(p) = \hat{b}_0^2 E[e^2(n)] = \hat{b}_0^2$ (will always decrease with p, look for "knee")

Model order estimation: minimize $AIC(p) = N \log \hat{\varepsilon}(p) + 2p$ or $MDL(p) = N \log \hat{\varepsilon}(p) + p \log N$
where the estimated model error is $\hat{\varepsilon}(p) = \hat{b}_0^2 = \hat{r}_x(0) + \sum_{k=1}^p \hat{a}_k \hat{r}_x(k)$

MA(q) model: moving average model with q zeroes



MA(q) (moving average)

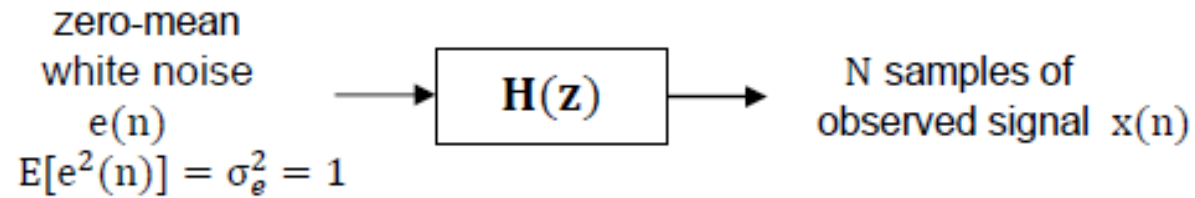
$$x(n) = \sum_{i=0}^q b_i e(n-i)$$

$$H(z) = \sum_{i=0}^q b_i z^{-i}$$

- Yule-Walker equations becomes non-linear in $\{\hat{b}_i\}_{i=1,\dots,q}$
- Methods exist to estimate $\{\hat{b}_i\}_{i=1,\dots,q}$ from $x(n)$ directly or from $\hat{r}_x(k)$
MATLAB: **armax**
- However: since the MA(q) model corresponds to a $q+1$ tap FIR-filter
 $r_x(k) = 0, |k| > q$
- Therefore an equivalent and simpler approach is to use the (non-parametric)
Blackman-Tukey estimate with a rectangular window of width $2q+1$:

$$\hat{P}_x(f) = \sum_{k=-q}^q \hat{r}_x(k) e^{-j2\pi \frac{f}{f_s} k}$$

ARMA(p,q) model: autoregressive moving average model with p poles and q zeroes



- Yule-Walker equations becomes non-linear in $\{\hat{a}_i\}_{i=1,\dots,p}$, $\{\hat{b}_i\}_{i=1,\dots,q}$
 - Methods exist to estimate $\{\hat{a}_i\}_{i=1,\dots,p}$, $\{\hat{b}_i\}_{i=1,\dots,q}$ from $x(n)$ directly or from $\hat{r}_x(k)$
- MATLAB: **armax**

- However a simpler three-step approach can be used:
- First, find $\{\hat{a}_i\}_{i=1,\dots,p}$ from MYWE, Modified Yule-Walker equations

$$\begin{bmatrix} \hat{r}_x(q) & \cdots & \hat{r}_x(q - (p - 1)) \\ \vdots & \ddots & \vdots \\ \hat{r}_x(q + (p - 1)) & \cdots & \hat{r}_x(q) \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \vdots \\ \hat{a}_p \end{bmatrix} = - \begin{bmatrix} \hat{r}_x(q + 1) \\ \vdots \\ \hat{r}_x(q + p) \end{bmatrix}$$

This gives $\hat{P}_{x,\text{denom}}(f) = \left| 1 + \sum_{i=1}^p \hat{a}_i e^{-j2\pi \frac{f}{f_s} i} \right|^2$

- Secondly, filter $x(n)$ with $G(z) = \text{denom}\{H(z)\} = 1 + \sum_{i=1}^p a_i z^{-i}$ giving a MA(q) signal, $y(n)$
 - Thus, finally, estimate $\hat{r}_y(k)$ and find $\hat{P}_{x,\text{nom}}(f) = \sum_{k=-q}^q \hat{r}_y(k) e^{-j2\pi \frac{f}{f_s} k}$ as for MA(q)
- Then $\hat{P}_x(f) = \frac{\hat{P}_{x,\text{nom}}(f)}{\hat{P}_{x,\text{denom}}(f)}$